



• 100% FREE · Live Cohort

👋 Hey beginners — this one's for you

Get started with Cloud & AI on AWS in just 5 days.

A friendly, **beginner-friendly intro** to AWS — no prior experience needed. Perfect for students, graduates, and anyone curious about cloud or AI.

- ✓ **Learn the essentials** — cloud, storage, compute, AI services
- ✓ **Hands-on demo on Day 5** — see AWS services working together
- ✓ **10 hours total** · live · 3 timezones to choose from

★★★★★

"Concepts are explained in a simple, step-by-step manner. Even advanced AWS topics feel clear and manageable."

Gururaj Rathod · DevOps Engineer

★★★★★

"I still go back to it as a reference. So much content — theory plus hands-on with real-world examples."

Antonio Villarán López · Cloud Architect

LIVE DATES

May 25 – 29

Mon to Fri · 2026 · 2 hrs/day

PICK YOUR TIMEZONE

EST 9:00 – 11:00 AM

GMT 2:00 – 4:00 PM

IST 6:30 – 8:30 PM

Save your seat at

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→ Limited live spots



FOR BEGINNERS · NO EXPERIENCE NEEDED

CLOUD & AI ON AWS



DAY 2 · S3 & CloudFront

MAY 25-29 · 2 HRS/DAY · • LIVE

Agenda

Day 1 ● What is cloud computing, AWS and Identity & access (IAM)



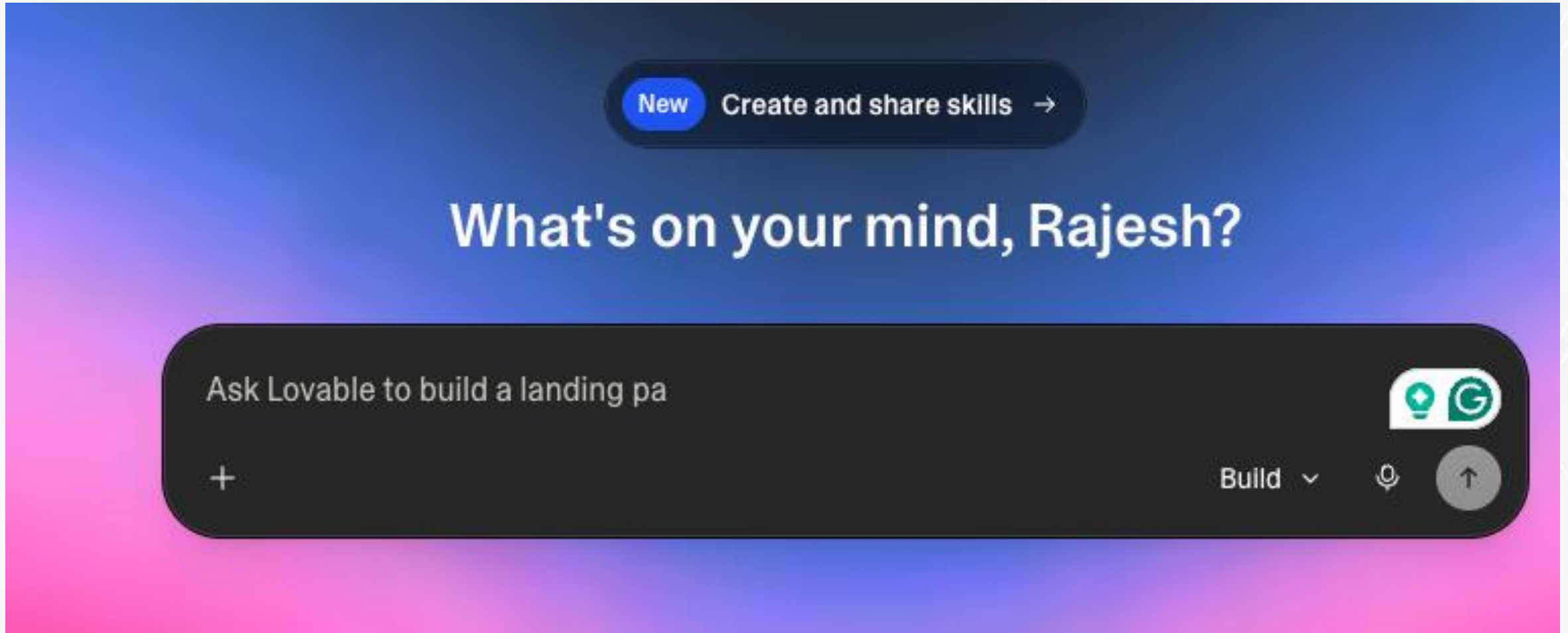
Day 2 ● Introduction to Storage (Amazon S3), and content delivery (AWS CloudFront)

Day 3 ● Compute (EC2, Lambda, ECS) and how services communicate (SQS, SNS)

Day 4 ● A friendly intro to AWS AI services: Bedrock, Rekognition, Polly, Textract

Day 5 ● Hands-on demo: Lambda, Textract, Bedrock, S3 and CloudFront chained together

Recap on Lovable.dev





Amazon S3 and AWS CloudFront

Storage and Content Delivery on AWS

Three ways to store data in the cloud

Same question — where does the data live — but three very different answers.



Object Storage

THE POSTAL WAREHOUSE

"You put parcels in and retrieve them by address. No folders, no hierarchy — just a flat warehouse where every item has a unique label and can be accessed via a URL."

Store any file, retrieve it by its key over HTTP.
Scales infinitely, no management needed.

Amazon S3



Block Storage

THE RAW HARD DRIVE

"A blank hard drive attached directly to your server. The OS formats it however it likes and treats it exactly like a local disk — full read/write access at the block level."

Low-latency disk attached to a single EC2 instance. Formatted and managed by the OS.

Amazon EBS



File Storage

THE SHARED FILING ROOM

"A filing room that multiple people can walk into simultaneously. Everyone sees the same folders and files — a familiar hierarchy, shared across many machines at once."

A mountable file system shared across multiple servers simultaneously. Standard folders and paths.

Amazon EFS

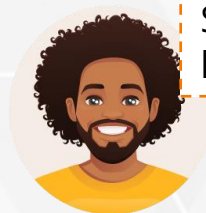
Which storage type should you use?

The right choice depends on how your application needs to access data.

	S3	EBS	EFS
Access method	URL / HTTP	Direct disk I/O	Network file path
Shared access	Yes — unlimited	One server only	Yes — many servers
Attach to EC2	Over internet	Directly attached	Mounted as drive
Max size	Unlimited	Up to 64 TB	Unlimited (elastic)
Best for	Files & media	OS disks / databases	Shared app files
AWS service	Amazon S3	Amazon EBS	Amazon EFS

What is Amazon S3

- **Global Object Storage** Solution – *Accessible globally but hosted regionally.*
- Data is stored as **objects** (with metadata and a globally unique identifier).
- Public service with multi-connect access.
- Use cases include **data lakes**, **websites**, **mobile apps**, **backups**, archives, and **big data analytics**.
- Designed to offer **99.999999999 (11 9s)** of durability.
- From a single CSV file to a high-resolution video.



Store 1 billion files → about 1 file lost every 10 years.

Object Storage – Not File or Block Storage



Amazon Simple Storage Service (Amazon S3)



Buckets are private by default



IAM Policies



Bucket Policies



ACLs

Access Methods



Important Note:



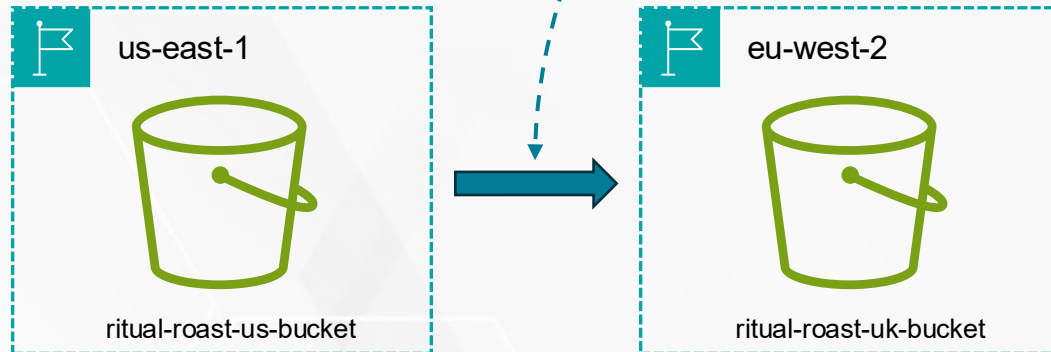
As of March 2026, [Amazon S3 introduces account regional namespaces](#) for general-purpose buckets, allowing buckets to be unique only within an account and region, rather than globally. New buckets can use a **<prefix>-<account-id>-<region>-an** format, eliminating the need to find globally unique names and preventing "bucket squatting"

What is Amazon S3 – Buckets and Objects

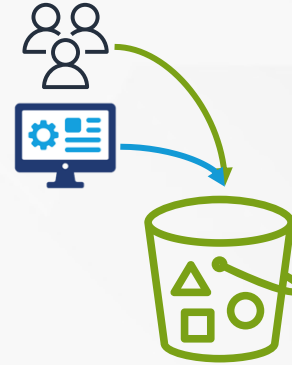
AWS never replicates your data to other regions unless you configure it to do so.

Increases Durability & Availability

Compliance & Data Sovereignty



- Globally unique names across AWS accounts
- 3 to 63 characters in length
- Lowercase letters, numbers, dots (.) and hyphens (-)
- Cannot start with an IP address
- 100 soft limit and 1000 hard limit per account



Objects

- Size from 0 bytes to 50TB
- Key: blueberry-muffin.jpg
- Value:



Flat File Structure

- blueberry-muffin-recipe.doc
- chocolate-muffin-recipe.doc
- lemon-drizzle-muffin-recipe.doc
- /archive/walnut-cake-v2-recipe.doc
- /latest/walnut-cake-v3-recipe.doc

Delimiter (/)

Prefix

The prefix and delimiter parameters can be used to limit the results of a list operation

AWS is constantly evolving...



<https://youtu.be/75mHfjPa8z0>



Create an Amazon S3 Bucket

Features and Configuration

S3 Storage Classes

All classes store the same data — the difference is retrieval speed, cost and minimum commitment.

Standard	Standard-IA	Intelligent-Tiering	One Zone-IA
SPEED Milliseconds	SPEED Milliseconds	SPEED Milliseconds	SPEED Milliseconds
COST Highest	COST Lower + retrieval fee	COST Auto-optimised	COST Lower (single AZ)
USE Active apps, websites, frequent access	USE Backups, DR copies, infrequent access	USE Unknown or unpredictable access patterns	USE Re-creatable data, secondary backups
MIN DAYS None	MIN DAYS 30 days	MIN DAYS None	MIN DAYS 30 days

Glacier Instant	Glacier Flexible	Glacier Deep Archive
SPEED Milliseconds	SPEED Mins – hours	SPEED Up to 12 hrs
COST Low	COST Very low	COST Lowest
USE Medical images, news media archives	USE Disaster recovery, long-term backups	USE Compliance archives, rarely accessed
MIN DAYS 90 days	MIN DAYS 90 days	MIN DAYS 180 days

S3 Key Features



Versioning

PROTECT YOUR DATA

"Every time you overwrite or delete a file, S3 keeps the previous version. You can restore any earlier copy at any time."

- Enabled per bucket — off by default
- Every PUT creates a new version ID
- Deleted objects get a delete marker — not permanently removed
- Restore any previous version instantly

Great safety net for accidental overwrites or deletes



Replication

COPY ACROSS REGIONS OR ACCOUNTS

"Automatically copies every new object to one or more destination buckets — in the same region, a different region, or even a different AWS account."

- Cross-Region (CRR) — resilience & latency
- Same-Region (SRR) — logs & compliance
- Versioning must be on in both buckets
- Objects keep same key and metadata

Commonly used for disaster recovery and data sovereignty



Static Website Hosting

SERVE A WEBSITE FROM S3

"Point S3 at your HTML, CSS and JS files and it serves them directly to browsers — no server, no EC2, no maintenance. Just upload and share the URL."

- Enable in bucket settings, set index.html
- AWS provides a public website URL
- Add CloudFront for HTTPS + custom domain
- Great for portfolios, docs, SPAs

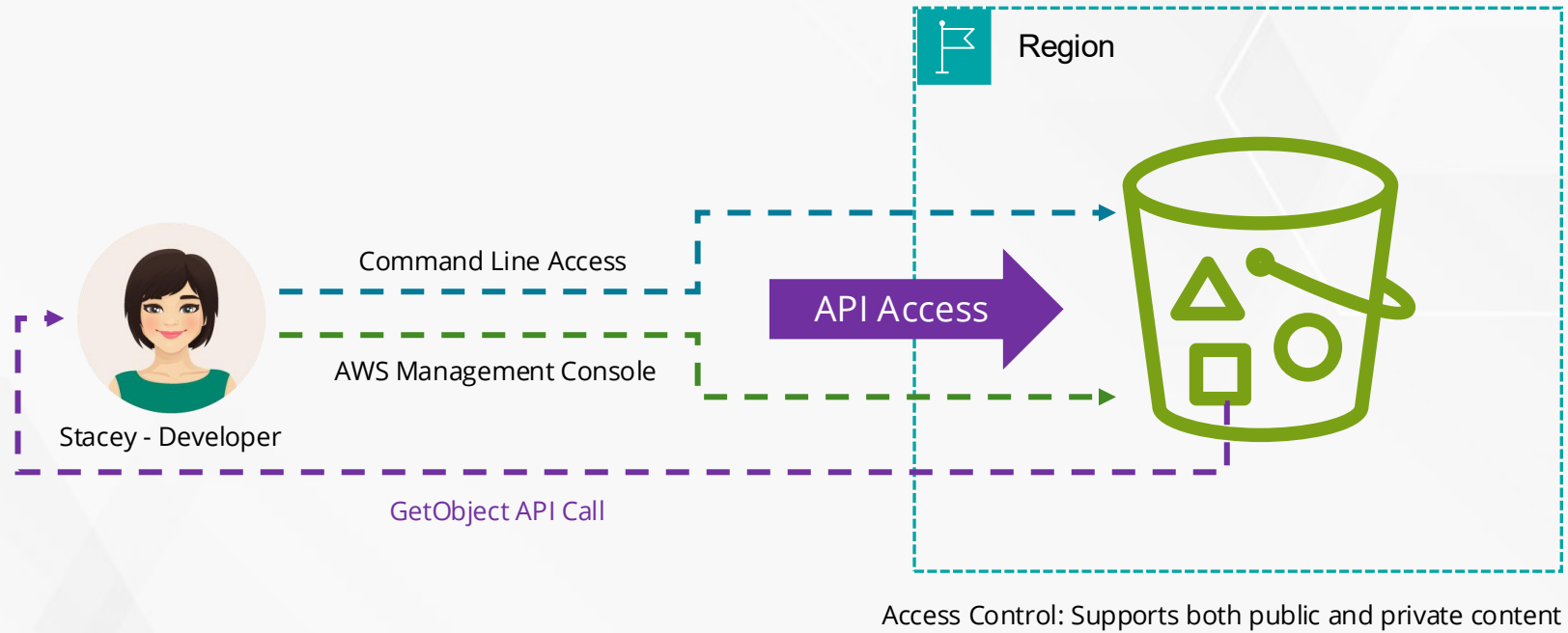
Cheapest way to host a website — often just cents per month



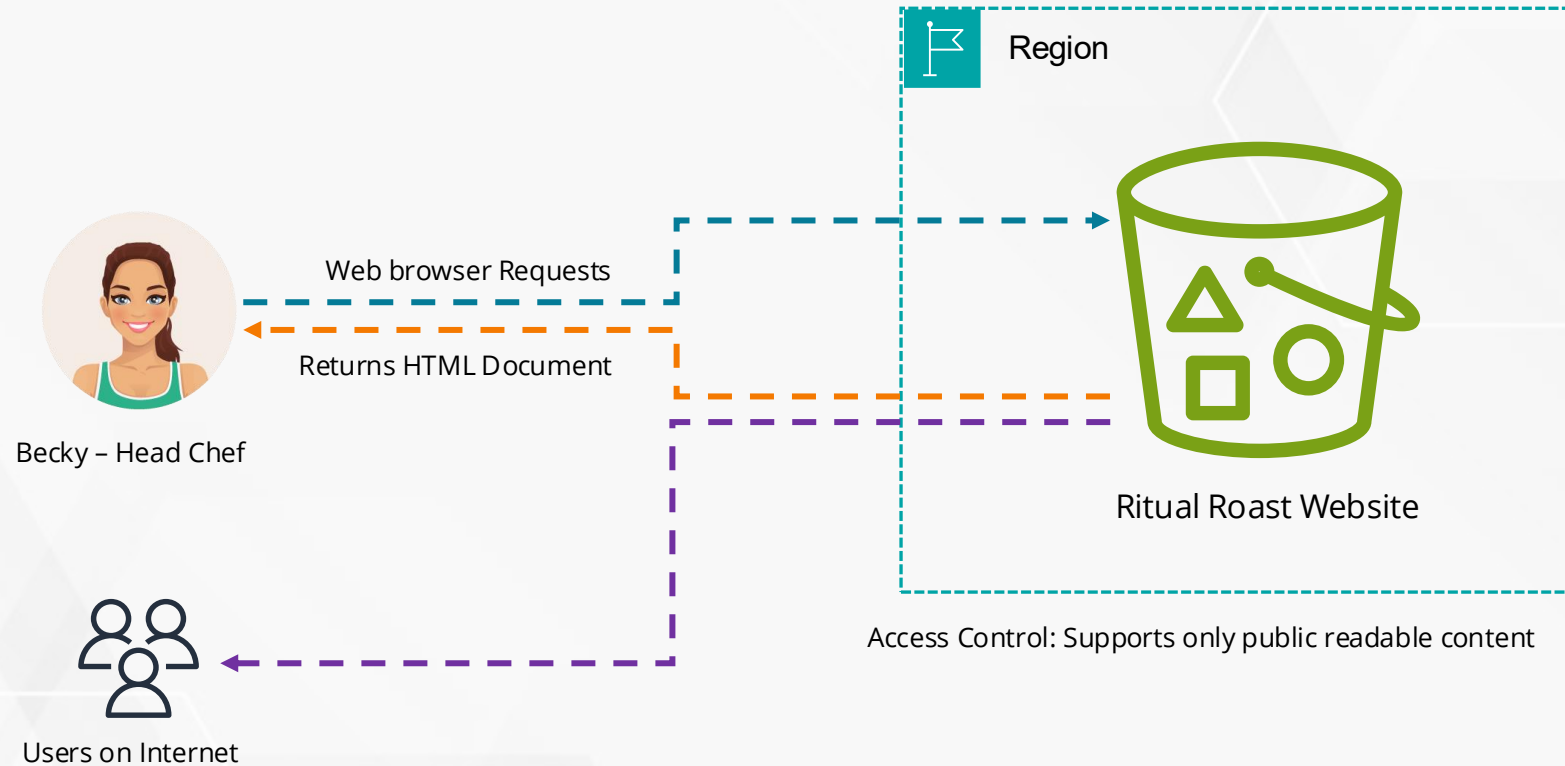
Amazon S3 Static Website Hosting

Hosting Front-End Static Websites

Accessing Amazon S3 Buckets – API Access



Accessing Amazon S3 Buckets – Website Endpoint



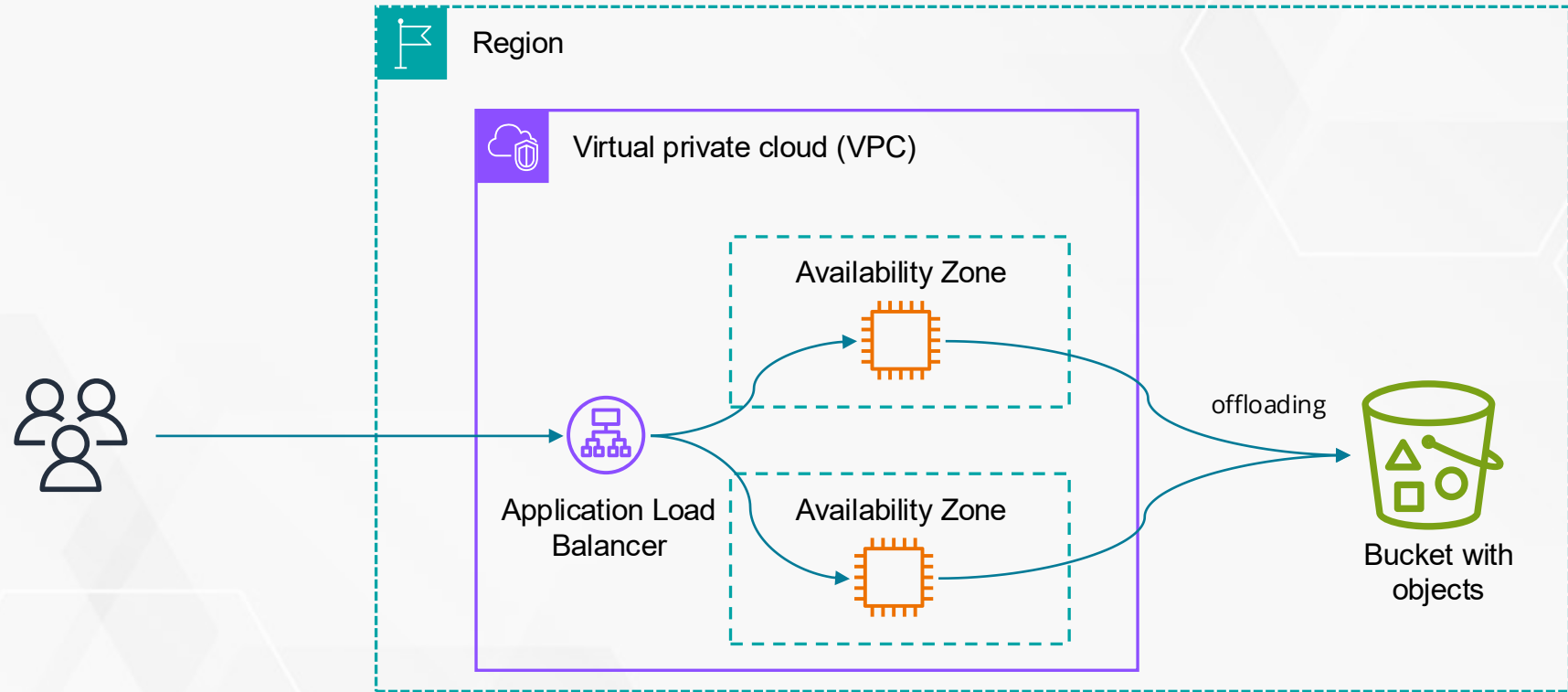
URL Format - Region Dependent

s3-website dash (-) Region - `http://bucket-name.s3-website.Region.amazonaws.com`
s3-website dot (.) Region - `http://bucket-name.s3-website-Region.amazonaws.com`

Static Website Hosting – Product Launch



Static Website Hosting – Offloading



Static Website Hosting

To configure a bucket to host a static website, follow these steps:

- Create a bucket with the name that will be your website hostname
- Upload static files to the bucket
 - You will need an **index.html** file
 - You will need an **error.html** file
- Make all files public or grant everyone read access
- Enable the bucket for Static Website Hosting
- Optionally configure a DNS CNAME to point to the Amazon URL. However, your **bucket name must match the custom domain name.**

What is Amazon CloudFront



What is an Edge Location?

A data centre in a major city that caches copies of your content. When a user requests a file, it is served from the nearest edge — not your origin server.

How it works

- 1 User requests a file (image, video, page)
- 2 CloudFront checks the nearest edge for a cached copy
- 3 Cache HIT: served instantly from edge. Cache MISS: fetched from your origin (S3, EC2), cached for next time

600+

Edge Locations

90+

Cities globally

47+

Countries



Speed

Millisecond-level latency. Users get content from a server near them, not halfway across the world.



Security

Built-in DDoS protection via AWS Shield. HTTPS by default. Works with WAF for extra control.



Origin Support

Works with S3, EC2, load balancers, or any HTTP server — on AWS or elsewhere.



Custom Domains

Attach your own domain and SSL certificate. Your users see `yoursite.com`, not an AWS URL.

CloudFront manages edge locations automatically — point it at your origin and it handles the rest. Used in the Day 5 capstone.



Amazon CloudFront

Creating a content delivery network for
your web application

Amazon CloudFront

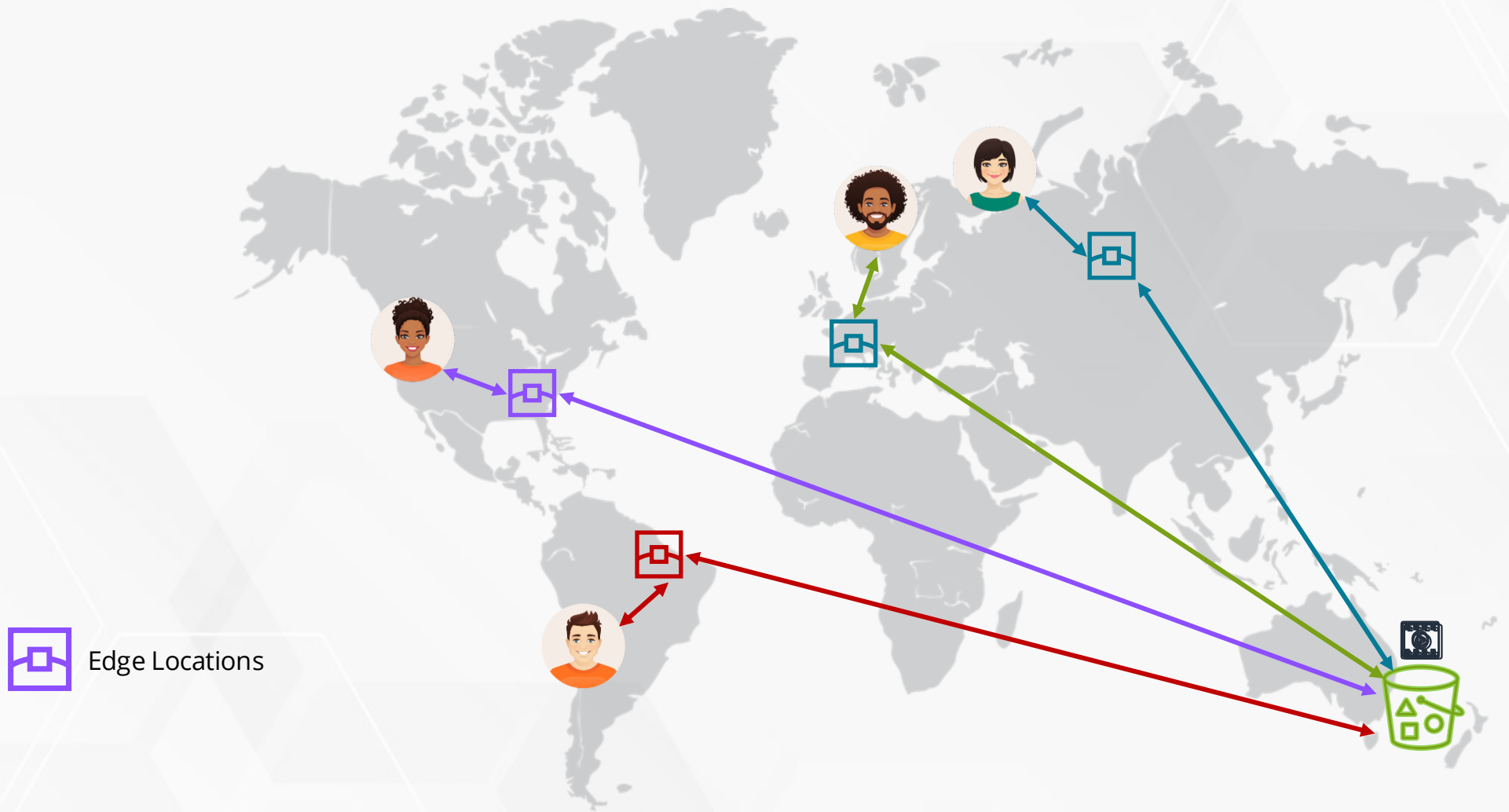


Customers may experience high latency issues

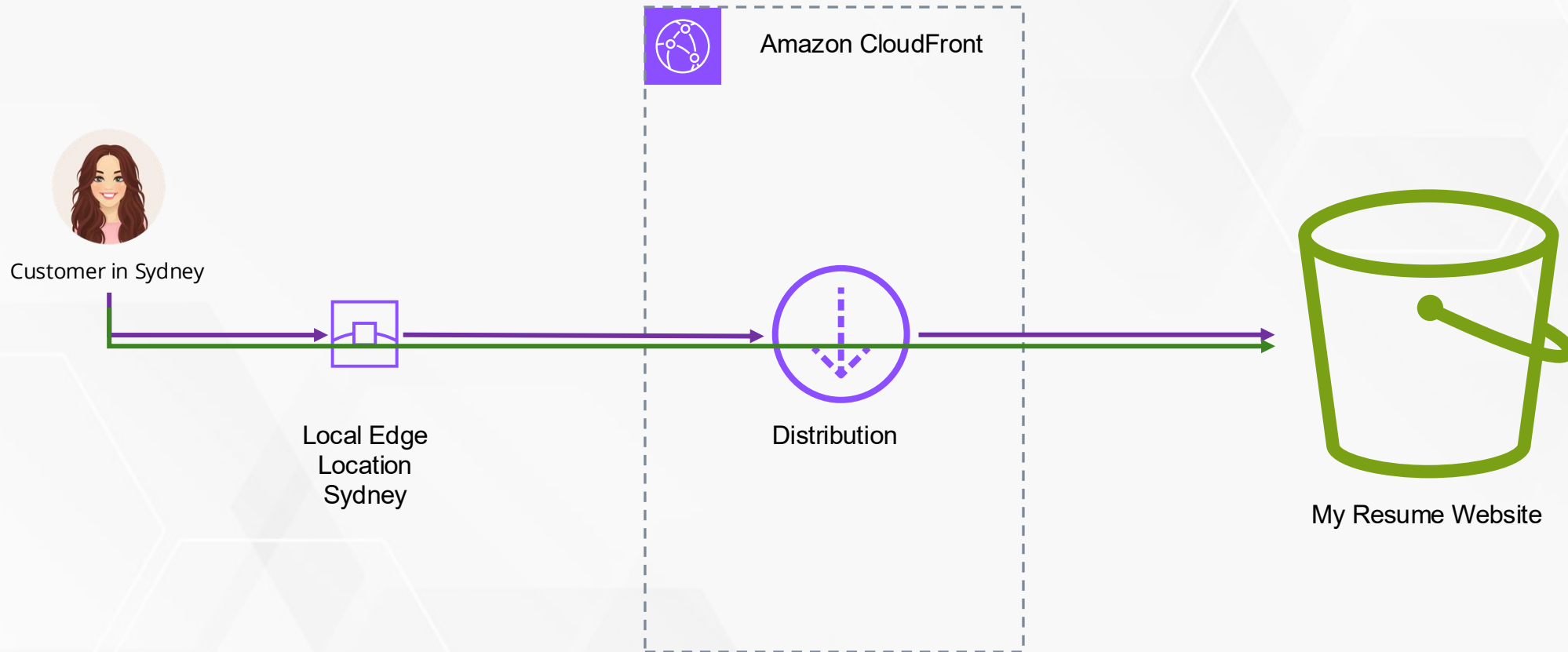


Repeat requests must be fetched from origin

Amazon CloudFront



Amazon CloudFront – myresume website





Create an Amazon CloudFront Distribution

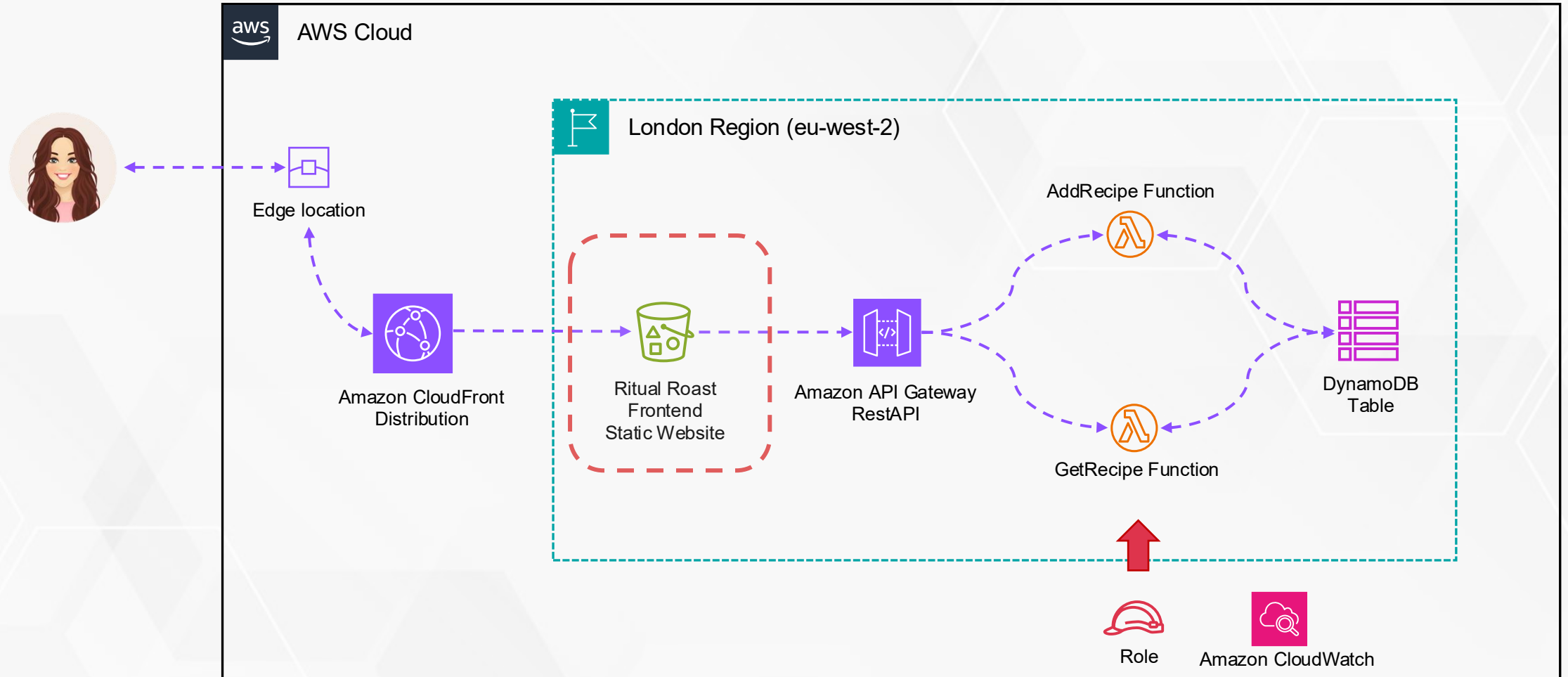
Designing a CDN for your web application



Amazon S3 & CloudFront

Real-World Architecture

Real-World Solution





Homework

- Create your Amazon S3 Bucket
- Upload the sample resume document to the bucket
- Create a CloudFront Distribution for the Amazon S3 bucket

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Scan any code to connect

